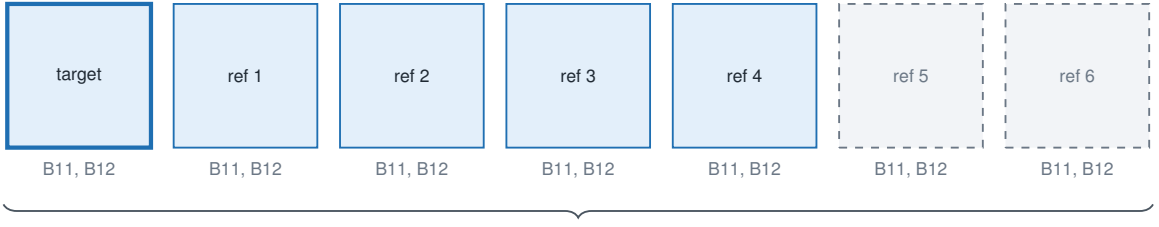
One tensor shape feeds every model

Level 05 - tensor contract. Target and references share one tile; missing references are padded and masked, never dropped.

Input x : (B, 7, 2, 100, 100)



ref_valid : (B, 7), bool

Example above: a fine-tuning sample has 4 real references, so slots 5 and 6 are padded and marked 0. Synthetic samples may have up to 6.

Output: logits (B, 1, 100, 100)
Same resolution as the input. Sigmoid and a threshold give the predicted mask.

Label: mask (B, 1, 100, 100)
Synthetic: $\Delta XCH_4 > 150$ ppb. Fine-tune: real mask; the noisy MBMP map is not used as the loss target.

Scale
20 m pixels (2×2 mean of 10 m chips), 100 × 100 = 2 km. A 10 km granule is a 5 × 5 grid of 25 chips.

Every model in the zoo accepts exactly this input and returns exactly this output; a change here is a change to the benchmark, so it needs an ADR and a version bump in the dataset card.